

Advanced Statistical Methods II

Lecture 10: Nonparametric Estimation of Survival Curve

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Outline of Today's Lecture

Why nonparametric survival estimation?

Risk sets and event times

Product-limit idea

Kaplan–Meier estimator

Nelson–Aalen estimator

Survival curves and median survival

KM plots for parametric goodness-of-fit

Uncertainty and practical issues

Summary

- ▶ Event time: T .

- Survival function: $S(t) = \mathbb{P}(T > t)$.

- Censoring time: C .

- Observed data: $Y = \min(T, C)$, $\delta = \mathbb{I}(T \leq C)$.

- Likelihood contribution: $f(y_i)^{\delta_i} S(y_i)^{1-\delta_i}$, under independent censoring.

Today we do **not** assume exponential, Weibull, gamma, or any specific parametric family. We estimate $S(t)$ directly from censored data.

The central question



Suppose we observe

$$(y_1, \delta_1), \dots, (y_n, \delta_n),$$

where $\delta_i = 1$ means an event was observed and $\delta_i = 0$ means right censoring.

How can we estimate the whole curve

$$S(t) = \mathbb{P}(T > t)$$

without specifying a distribution for T ?

- ▶ We cannot simply treat censored times as event times.
- ▶ We cannot simply discard censored observations.
- ▶ We need an estimator that uses the information: *subject survived at least up to the censoring time*.

Why not just use the empirical survival curve?

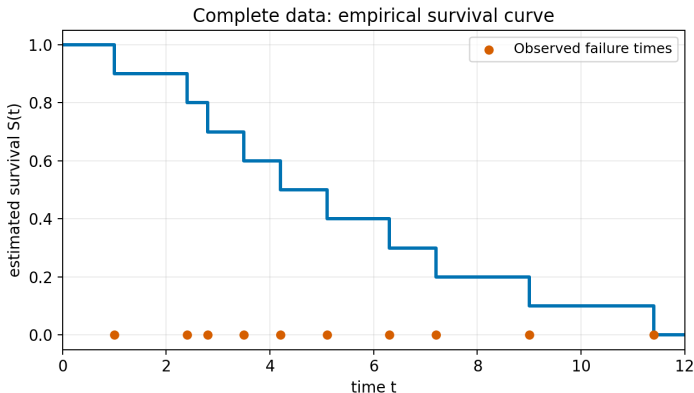


If we observed complete failure times $t_1, \dots, t_n$, a natural estimator would be

$$\hat{S}_{\text{emp}}(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(t_i > t) = \frac{\#\{i : t_i > t\}}{n}.$$

This empirical survival estimator is the starting point for the nonparametric discussion; see, for example, the complete-data derivation in [Wang \(2005, pp. 27–30\)](#).

Complete data: empirical survival curve



With complete data, every observed time that is an event causes a jump down. With censoring, the main difficulty is deciding **who should still be counted in the denominator** at each event time.

δ_i	What we observe	What we know about T_i
1	event at y_i	$T_i = y_i$
0	censored at y_i	$T_i > y_i$

This is why Kaplan–Meier estimation is built around **risk sets** and **conditional survival probabilities**.

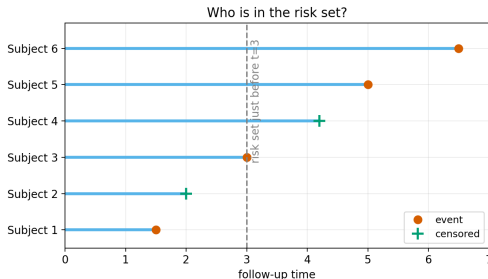
$$y_{(1)} < y_{(2)} < \cdots < y_{(k)}$$

For the event time $y_{(j)}$, define

$$R_j = \{i : Y_i \geq y_{(j)}\}, \quad N_j = |R_j|, \quad d_j = \#\{i : Y_i = y_{(j)}, \delta_i = 1\}.$$

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The risk set just before time $y_{(j)}$ contains individuals who have not yet failed and have not yet been censored before $y_{(j)}$.



At an event time, the estimated conditional failure probability is $\hat{h}_j = \frac{d_j}{N_j}$. Thus the estimated conditional survival probability through that event time is

$$1 - \hat{h}_j = 1 - \frac{d_j}{N_j}.$$

Who is in the risk set?



Suppose the next event time is $t = 3$.

- ▶ A subject who failed before 3 is not in the risk set.
- ▶ A subject censored before 3 is not in the risk set.
- ▶ A subject censored after 3 is in the risk set at 3.
- ▶ A subject failing at 3 is in the risk set just before 3.

Interpretation

The denominator N_j is not the original sample size. It is the number of individuals who were still **under observation and event-free** just before the event time.

For ordered event times, survival beyond a later time can be decomposed into a product of conditional survival probabilities.

For example,

$$\mathbb{P}(T > t_3) = \mathbb{P}(T > t_1)\mathbb{P}(T > t_2 \mid T > t_1)\mathbb{P}(T > t_3 \mid T > t_2).$$

More generally, if $y_{(j)} \leq t$, estimate each conditional factor by

$$\hat{\mathbb{P}}(\text{survive past } y_{(j)} \mid \text{at risk just before } y_{(j)}) = 1 - \frac{d_j}{N_j}.$$

The Kaplan–Meier estimator multiplies these factors.

Why multiplication?



At each event time, we estimate a **conditional** probability.

$$\hat{S}(t) = \hat{\mathbb{P}}(T > y_{(1)}) \times \hat{\mathbb{P}}(T > y_{(2)} \mid T > y_{(1)}) \times \cdots .$$

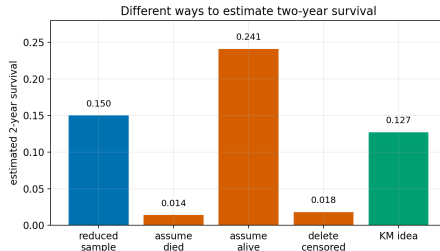
The estimator is called a **product-limit estimator** because the survival curve is built as a product of many local survival proportions.

A two-year survival example

A study has two cohorts.

- ▶ 100 patients recruited in year 0 and followed for 2 years.
- ▶ 1000 patients recruited in year 1 and followed for 1 year before the study ends.
- ▶ Deaths: 70 in year 1 from cohort 1, 15 in year 2 from cohort 1, and 750 in year 1 from cohort 2.

Naive approaches can give very different answers depending on how censored patients are handled.



Two-year example: product-limit calculation



The product-limit idea estimates

$$S(2) = \mathbb{P}(T > 2) = \mathbb{P}(T > 1)\mathbb{P}(T > 2 \mid T > 1).$$

The Kaplan–Meier-type calculation is

$$\hat{\mathbb{P}}(T > 1) = \frac{30 + 250}{1100} = \frac{280}{1100},$$

where 30 from cohort 1 and 250 from cohort 2 survive year 1.

Then

$$\hat{\mathbb{P}}(T > 2 \mid T > 1) = \frac{15}{30}.$$

Hence

$$\hat{S}(2) = \frac{280}{1100} \times \frac{15}{30} = 0.127.$$

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What does each factor mean?



At event time $y_{(j)}$,

$$\frac{d_j}{N_j}$$

is the estimated conditional probability of failure at that time among those still at risk.

Therefore,

$$1 - \frac{d_j}{N_j}$$

is the estimated conditional probability of surviving through that event time.

The KM curve falls only at uncensored event times. It stays flat at censoring times, but censoring changes later risk sets.

Small worked example



Consider the data

3, 2⁺, 0, 1, 5⁺, 3, 5,

where + denotes right censoring.

The uncensored event times are

0, 1, 3, 5.

Event time $y_{(j)}$	N_j	d_j	factor $1 - d_j/N_j$
0	7	1	6/7
1	6	1	5/6
3	4	2	1/2
5	2	1	1/2

Small worked example: calculations



Using

$$\hat{S}(t) = \prod_{y_{(j)} \leq t} \left(1 - \frac{d_j}{N_j}\right),$$

we get

$$\hat{S}(0) = 1 - \frac{1}{7} = \frac{6}{7} = 0.86,$$

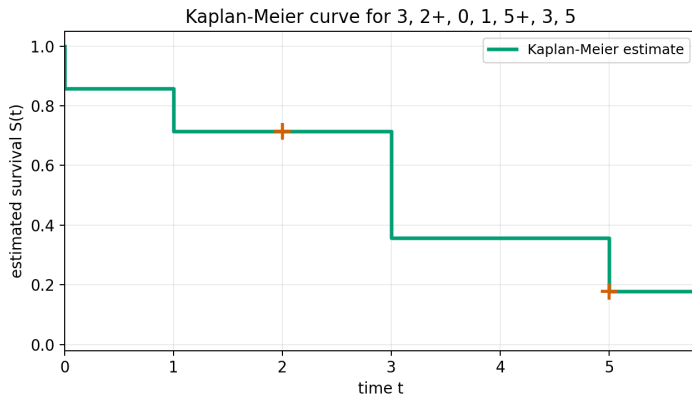
$$\hat{S}(1) = \frac{6}{7} \left(1 - \frac{1}{6}\right) = \frac{5}{7} = 0.71,$$

$$\hat{S}(3) = \frac{5}{7} \left(1 - \frac{2}{4}\right) = \frac{5}{14} = 0.36,$$

$$\hat{S}(5) = \frac{5}{14} \left(1 - \frac{1}{2}\right) = \frac{5}{28} = 0.18.$$

At time 3 there are two failures, so the drop is larger: multiply by $1 - 2/4 = 1/2$.

Small worked example: the KM curve



- ▶ The curve is a step function.
- ▶ It drops at 0, 1, 3, 5, the event times.
- ▶ The censoring marks at 2 and 5 do not themselves create a drop.

Suppose several events occur at the same observed time $y_{(j)}$.

- ▶ Count all failures at that time in d_j .
- ▶ Use one multiplicative factor:

$$1 - \frac{d_j}{N_j}.$$

- ▶ Censored observations at the same recorded time are usually treated as occurring just after failures at that time, unless a study-specific convention says otherwise.

In the small example

At time 3, two individuals fail and $N_j = 4$, so the factor is

$$1 - \frac{2}{4} = \frac{1}{2}.$$

$$L \propto \prod_{i=1}^n f(y_i)^{\delta_i} S(y_i)^{1-\delta_i}.$$

So the KM estimator is not just a heuristic curve. It also arises from a likelihood principle.

A likelihood intuition for KM



At event time $y_{(j)}$, think of the discrete hazard

$$h_j = \mathbb{P}(T = y_{(j)} \mid T \geq y_{(j)}).$$

Among N_j people at risk, d_j fail at that time. The local likelihood contribution behaves like

$$h_j^{d_j} (1 - h_j)^{N_j - d_j}.$$

Maximizing this gives

$$\hat{h}_j = \frac{d_j}{N_j}.$$

Therefore the estimated survival through that event time is

$$1 - \hat{h}_j = 1 - \frac{d_j}{N_j}.$$

Multiplying these local survival factors gives the KM estimator.

Efron's redistribution-of-mass construction



The product-limit formula can also be understood through a constructive algorithm due to [Efron \(1967\)](#). A clear four-step version is discussed in [Sun \(n.d., Unit 5, pp. 5–6\)](#).

1. Arrange the observations in increasing order; if there are ties, put failures before censored observations.
2. Put mass $1/n$ at each observation.
3. Move from left to right. Whenever a censored observation is reached, redistribute its current mass equally among observations to its right.
4. Continue until all censoring masses have been redistributed. If the largest observation is censored, its mass remains beyond the largest observed time.

Censoring does not destroy probability mass. It says: “the failure happened somewhere later.” Efron’s algorithm literally moves that mass to the right.

Nelson–Aalen estimator

Recall the cumulative hazard function

$$H(t) = \int_0^t h(u) du = -\log S(t).$$

At event time v_j , the estimated discrete hazard is

$$\hat{h}_j = \frac{d_j}{Y(v_j)}.$$

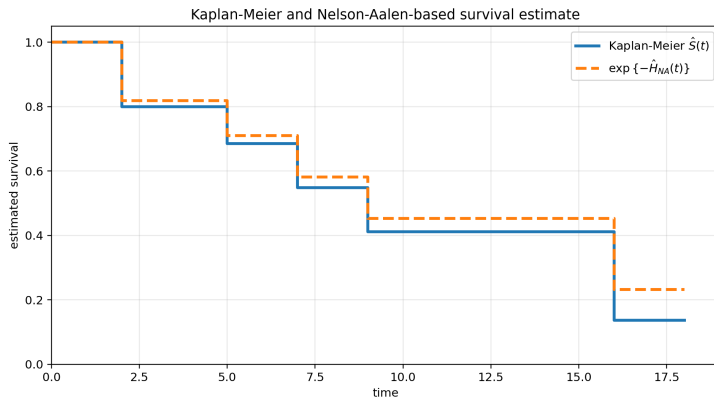
The Nelson–Aalen estimator of cumulative hazard is

$$\hat{H}_{NA}(t) = \sum_{j: v_j \leq t} \frac{d_j}{Y(v_j)}.$$

A survival estimate based on it is

$$\hat{S}_{NA}(t) = \exp\{-\hat{H}_{NA}(t)\}.$$

KM and Nelson–Aalen on the same data



- ▶ KM multiplies local survival factors: $\prod (1 - d_j/Y_j)$.
- ▶ Nelson–Aalen adds local hazard increments: $\sum d_j/Y_j$.
- ▶ The two are close when each d_j/Y_j is small.

- $\hat{S}(t) = 0.80$: estimated 80% surviving beyond time t .

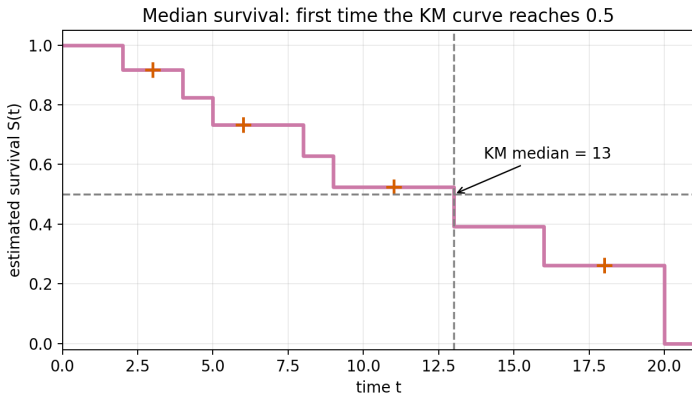
- ▶ Vertical drops: observed event times.
- ▶ Larger drops: more failures or a smaller risk set.
- ▶ Censoring marks: individuals leaving observation without event.
- ▶ Flat segments: no event observed during that interval.

The right tail of a KM curve can be unstable when very few subjects remain at risk.

The Kaplan–Meier median is the first time at which the estimated survival curve reaches or goes below 0.5:

$$\hat{m} = \inf\{t : \hat{S}(t) \leq 0.5\}.$$

Median survival: graphical illustration



Using KM for parametric goodness-of-fit



The Kaplan–Meier estimator is nonparametric, so it can be used as a diagnostic reference for a proposed parametric model.

1. Fit or hypothesize a parametric family for T .
2. Find a transformation of $S(t)$, $F(t)$, or $H(t)$ that should look simple under that family.
3. Replace $S(t)$ by $\hat{S}(t)$, usually the KM estimator.
4. Check the transformed plot visually.

This is a graphical goodness-of-fit diagnostic. It can reveal serious mismatch, but it is not by itself a formal test.

The idea is discussed, for example, in [Sun \(n.d., Unit 5, pp. 9–10\)](#).

GOF idea 1: exponential survival

Suppose

$$T \sim \text{Exponential}(\lambda).$$

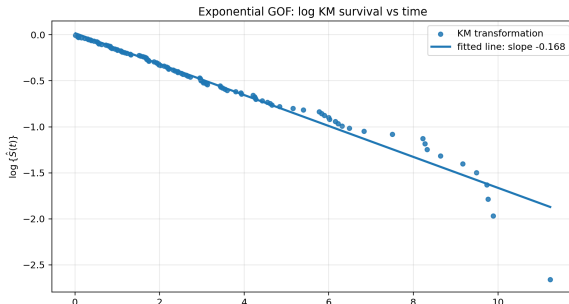
Then

$$S(t) = \exp(-\lambda t), \quad \log S(t) = -\lambda t.$$

Therefore, under an exponential model,

$\log\{\hat{S}(t)\}$ plotted against t

should look approximately linear with negative slope.



GOF idea 2: Weibull survival



For a Weibull model, write

$$S(t) = \exp\{-(\lambda t)^p\}.$$

Then

$$-\log S(t) = (\lambda t)^p,$$

and hence

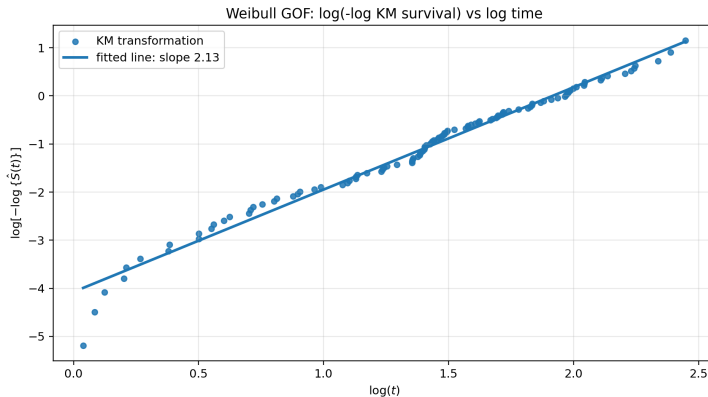
$$\log[-\log S(t)] = p \log \lambda + p \log t.$$

Therefore, under a Weibull model,

$$\log[-\log\{\hat{S}(t)\}] \quad \text{plotted against} \quad \log t$$

should look approximately linear.

Weibull GOF plot



- If the plot is roughly straight, a Weibull assumption is plausible.
- The slope estimates the Weibull shape parameter p , at least heuristically.
- Strong curvature suggests the Weibull family may not be adequate.

A brief look at uncertainty: Greenwood's formula



A common large-sample variance estimate for the Kaplan–Meier estimator is Greenwood's formula:

$$\widehat{\text{Var}}\{\widehat{S}(t)\} \approx \widehat{S}(t)^2 \sum_{j: y_{(j)} \leq t} \frac{d_j}{N_j(N_j - d_j)}.$$

- ▶ The sum accumulates uncertainty over event times.
- ▶ The variance increases when risk sets become small.
- ▶ This motivates confidence intervals and confidence bands for survival curves.

Greenwood's formula and its heuristic derivation are discussed, for example, in [Wang \(2005, pp. 42–44\)](#).

Why risk sets control uncertainty



At a given event time $y_{(j)}$, the local failure count d_j is like a binomial count among N_j individuals at risk.

$$d_j \approx \text{Binomial}(N_j, h_j).$$

- ▶ Large N_j : stable estimate of the local failure probability.
- ▶ Small N_j : unstable local estimate.
- ▶ Late event times often have small risk sets, so tail estimates are more variable.

The KM curve is usually most reliable where a reasonable number of subjects remain at risk.

- ▶ number of individuals at risk at selected times;
- ▶ number of events;
- ▶ number censored;
- ▶ median survival, if estimable;
- ▶ confidence intervals for survival probabilities or median survival;
- ▶ whether censoring is plausibly independent or non-informative.

The Kaplan–Meier estimator assumes that censoring does not carry extra information about the event time after conditioning on the design/covariates being considered.

Key takeaway messages



1. The Kaplan–Meier estimator estimates $S(t) = \mathbb{P}(T > t)$ without specifying a parametric distribution.
2. The estimator is built from event times, risk sets, and event counts.
3. At each event time, the local survival factor is $1 - d_j/N_j$.
4. Censored observations do not create jumps but affect later risk sets.
5. The KM curve is a step function and the median survival is the first time it reaches or crosses 0.5.
6. The far right tail should be interpreted carefully, especially when few individuals remain at risk.

Question	Next tool
How do we compare two survival curves?	log-rank test
How do covariates affect survival?	Cox proportional hazards model
How do we model hazard ratios?	partial likelihood
How do we check proportional hazards?	residuals and diagnostics

The risk-set idea from today will reappear in the Cox model.

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